IMAGE SEGMENTATION

1. Difference between Image Classification vs. Image Segmentation vs. Object Detection

1. Image Classification

Objective:
To assign a single label to an entire image based on the primary object or scene depicted.

Characteristics:

Produces one class label per image.

Does not localize objects within the image.

Suitable for tasks where only presence matters (e.g., identifying if an X-ray contains signs of pneumonia).

Output Example:

Input: Image of a dog.

Output: "dog"

Use Cases:

Medical diagnosis from radiographs.

Content moderation.

2. Object Detection

Objective:
To identify and locate multiple objects within an image using bounding boxes, along with their respective class labels.

Characteristics:

- Outputs a set of bounding boxes with class labels.
- Enables localization of objects within an image.
- Handles multiple instances and categories simultaneously.

Output Example:

Input: Image with a dog and a car.

Output:
```
[
  {"label": "dog", "bbox": [x1, y1, x2, y2]},
  {"label": "car", "bbox": [x1, y1, x2, y2]}
]
```
Use Cases:

Autonomous driving (detecting pedestrians, vehicles, traffic signs).

Surveillance systems.

Retail analytics (customer tracking, product placement).

3. Image Segmentation

Objective:
To classify each pixel in the image, providing a more detailed understanding of object shapes and boundaries.

Types:

Semantic Segmentation: Labels all pixels belonging to the same class uniformly.

Instance Segmentation: Labels each object instance individually, even if they belong to the same class.

Characteristics:

- Delivers pixel-level annotations.
- Offers the highest granularity among the three.
- Requires more computational resources and precise annotations.

Output Example:

Input: Image with two dogs.

Output: Pixel mask where each dog is separately segmented.

Use Cases:

Medical imaging (organ/tumor segmentation).

Satellite image analysis.

Augmented reality.

Feature	Image Classification	Object Detection	Image Segmentation
Task Output	Single class label	Bounding boxes + labels	Pixel-wise class labels
Object Localization	Not supported	Bounding box	Pixel-level
Handles Multiple Objects	No	Yes	Yes
Output Complexity	Low	Medium	High
Example Use Case	Scene recognition	Pedestrian detection	Medical image analysis

2. What is the relation between input image size and output size of Image Segmentation Models?

In image segmentation, understanding how the input image size affects the output segmentation mask size is crucial for designing and training effective models. Here's a professional overview of the relationship:

Architecture Type	Output Size Compared to Input	Notes
U-Net, SegNet	Same	Encoder-decoder with upsampling
DeepLabV3+	Usually same	Uses dilated convolutions + upsampling
Basic CNN (no decoder)	Smaller	Size reduced due to strides and pooling

Custom models	Varies	Output size depends on layer configurations

3. What are segmentaton masks in segmentation task?

A **segmentation mask** is a 2D or 3D array (depending on the task type) where:

- **Each pixel value corresponds to a specific class label**, indicating which object or region that pixel belongs to.
- The mask provides a **pixel-wise annotation** of the image.

In image segmentation, the goal is to assign a **class label** to **every pixel** in the input image. The segmentation mask acts as the **ground truth (for training)** or the **predicted output (for inference)**.

4. Differentiate between DICE Score and IOU Score. What are the loss functions and evaulation metrics used for segmentation task?

DICE Score (F1 Score / Sørensen–Dice Coefficient)

- Measures the **overlap** between predicted and ground truth masks.
- Formula:
  Dice=2×∣A∩B∣∣A∣+∣B∣\text{Dice} = \frac{2 \times |A \cap B|}{|A| + |B|}Dice=∣A∣+∣B∣2×∣A∩B∣
- Where:
  - AAA: Predicted mask
  - BBB: Ground truth mask
- **Range**: 0 (no overlap) to 1 (perfect overlap)

IOU Score (Intersection over Union / Jaccard Index)

- Measures the **ratio** of intersection over union of the predicted and actual masks.

- Formula:
  IoU=∣A∩B∣∣A∪B∣\text{IoU} = \frac{|A \cap B|}{|A \cup B|}IoU=∣A∪B∣∣A∩B∣
- **Range**: 0 to 1 (higher is better)

DICE vs IOU: Key Differences

Aspect	DICE Score	IOU Score
Formula	(\frac{2	A \cap B
Sensitivity	More sensitive to small objects	Less sensitive to small overlaps
Interpretation	Similar to F1 score (harmonic mean)	Direct overlap ratio
Value Tendency	Usually higher than IoU	Slightly stricter (lower than Dice)